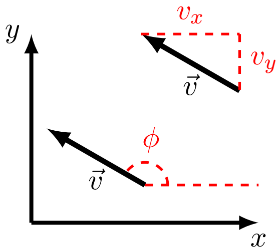
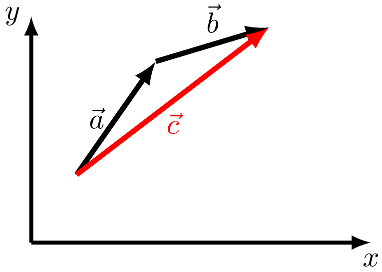
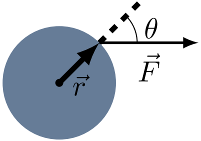


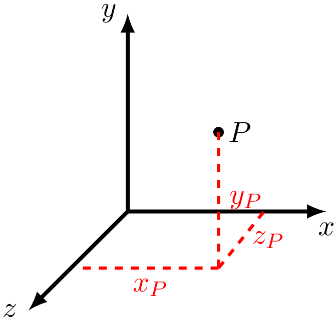
A vector in 2D can be specified by two Cartesian components (v_x, v_y) *or its length,* v*, and an angle with respect to the* x *direction,* ϕ*.*



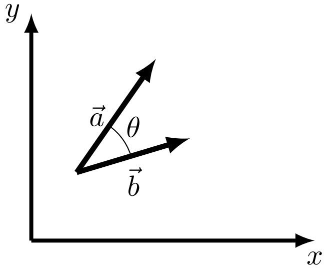
A graphical illustration of $\vec{c} = \vec{a} + \vec{b}$.



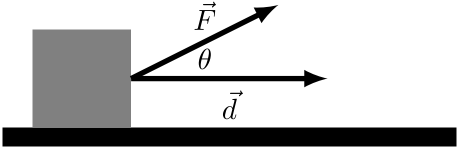
The magnitude of cross product between a force, $\vec{F}$*, and the position vector,* $\vec{r}$*, where the force is applied tells us the angular acceleration of an object:* $||\vec{r} \times \vec{F}|| = rF\sin\theta$*.*



*A **right-handed** Cartesian coordinate system, where $\hat{x} \times \hat{y} = \hat{z}$. ($\hat{z}$ is out of the page). The point P has coordinates (x_P, y_P, z_P).*



The scalar product between vectors depends on the angle between them when they are placed tail to tail.



The scalar product between force and displacement, $W = \vec{F} \cdot \vec{d} = F d \cos\theta$ *(Work) tells us about the change in speed.*